

11.Credit Score Classification using Machine Learning

Aim

To implement a **Credit Score Classification** program in Python using a machine learning algorithm and classify customers into different credit score categories.

Algorithm

1. Import the required libraries.
2. Create the dataset containing customer features and credit score classes.
3. Separate the input features and target class.
4. Split the dataset into training and testing data.
5. Train the classification model using the training data.
6. Predict the credit score class for the test data.
7. Calculate the accuracy of the model.
8. Display the actual and predicted results.

Program:

```
from sklearn.tree import DecisionTreeClassifier

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score

X = [

    [25, 30000, 1],

    [30, 40000, 2],

    [35, 50000, 3],

    [40, 60000, 4],

    [45, 70000, 5],

    [50, 80000, 6],

    [28, 35000, 1],

    [38, 55000, 3],

    [48, 75000, 5],

    [55, 90000, 6]
```

```
]
```

```
y = [
```

```
    "Poor",
```

```
    "Poor",
```

```
    "Average",
```

```
    "Average",
```

```
    "Good",
```

```
    "Good",
```

```
    "Poor",
```

```
    "Average",
```

```
    "Good",
```

```
    "Good"
```

```
]
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=42
```

```
)
```

```
model = DecisionTreeClassifier(random_state=42)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Actual Credit Scores:", y_test)
```

```
print("Predicted Credit Scores:", y_pred)
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("Accuracy:", accuracy * 100, "%")
```

```
new_customer = [[32, 45000, 2]]

prediction = model.predict(new_customer)

print("New Customer Credit Score:", prediction[0])
```

OUTPUT:

```
    "Good"
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = DecisionTreeClassifier(random_state=42)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Credit Scores:", y_test)
print("Predicted Credit Scores:", y_pred)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy * 100, "%")

new_customer = [[32, 45000, 2]]
prediction = model.predict(new_customer)

print("New Customer Credit Score:", prediction[0])
```

▼ ... Actual Credit Scores: ['Good', 'Poor', 'Good']
Predicted Credit Scores: ['Good' 'Poor' 'Good']
Accuracy: 100.0 %
New Customer Credit Score: Average

RESULT:

Thus, the **Credit Score Classification** task was successfully implemented using a **Decision Tree Classifier** in Python, and the credit score category of a new customer was successfully predicted.